

Notes: Julio & Yook (JF, 2012)

Political Uncertainty and Corporate Investment Cycles

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1 Big picture

- **Question.** Do firms postpone investment when national elections create uncertainty about future policy, leadership, and the economic environment?
- **Core answer.** Yes. Across 48 countries from 1980 to 2005, corporate investment rates are significantly lower in election years than in nonelection years, even after controlling for Tobin's Q , cash flow, GDP growth, firm fixed effects, and year effects.
- **Main magnitude.** The baseline coefficient on the election-year dummy is about -0.0036 to -0.0041 in investment scaled by beginning-of-year assets. The authors interpret this as roughly a 4.8% reduction relative to normal investment rates in the year leading up to an election.
- **Mechanism.** The paper argues that the evidence is most consistent with a real-options channel: when election outcomes may change future taxes, regulation, spending, or macro policy, firms wait before committing irreversible capital.
- **Cross-sectional prediction.** If uncertainty is the channel, investment should fall more when elections are close, when policy discretion is high, when government is less stable, and when industries are politically sensitive. The paper finds evidence broadly consistent with these predictions.
- **Alternative explanations tested.** The authors examine election timing endogeneity, incumbent opportunism, political connections, business-cycle manipulation, large-country effects, financial crises, and random placebo elections.
- **Cash-holding evidence.** Firms increase cash holdings during election years by about 0.0053, almost offsetting the investment decline of 0.0065. This supports the idea that firms delay rather than permanently destroy planned investment.
- **Takeaway.** Political uncertainty is not only a macro variable; it directly shapes corporate investment cycles. Elections can generate temporary real distortions by encouraging firms to wait.

1.1 Incremental contribution of the paper

- **Moves election-cycle analysis from macro aggregates to firm investment.** Earlier political business cycle work focuses on GDP, money growth, inflation, government spending, or aggregate investment. This paper uses firm-level investment to show that the political process affects real corporate decisions.
- **Builds a large international election-firm panel.** The paper manually classifies national elections, political systems, election timing, and election outcomes for 48 countries and merges them to Worldscope firm-year data.
- **Distinguishes uncertainty from opportunistic policy manipulation.** The paper does not merely show that investment is lower before elections; it shows that the decline is stronger when outcomes are close and policy discretion is high, while political business cycle variables show little systematic manipulation.
- **Connects political uncertainty to real-options theory.** The result is interpreted as a waiting effect: when capital is partly irreversible, greater uncertainty raises the option value of delay.
- **Documents cash substitution.** The paper adds cash-holding evidence showing that the foregone investment appears to be temporarily held as cash, consistent with delayed investment rather than general financial distress.

2 Data and measurement

2.1 Election data and political systems

- The election sample covers national elections in 48 countries between 1980 and 2005.

- The paper identifies the politically relevant election type by country. Presidential systems use presidential elections; parliamentary systems use legislative elections; hybrid systems are classified according to which office has greater executive authority.
- The final data include 31 countries with legislative elections, 16 countries with presidential elections, and Israel with prime-ministerial elections.
- Election timing is classified as fixed or flexible. Fixed timing is more plausibly exogenous because the election date is determined by electoral law rather than by the incumbent's strategic choice.
- The paper records whether elections lead to leadership or party changes, whether the incumbent is market-friendly or left-leaning, and how close the election is based on vote margins.

Interpretation: The political data are central because the identification strategy uses variation in when elections occur, whether their timing is plausibly exogenous, and how uncertain the electoral outcome is.

2.2 Firm-level investment data

- Firm data come from Thomson Financial's Worldscope database.
- The final sample contains 101,587 firm-year observations after dropping firms with missing investment, Tobin's Q , or cash-flow variables.
- The main dependent variable is the investment rate: capital expenditures scaled by beginning-of-year book assets.
- Key controls include Tobin's Q , cash flow, and country-level real GDP growth.
- Most specifications include firm fixed effects and year fixed effects; some include firm-by-continent/year effects to absorb region-specific shocks.

Interpretation: Firm fixed effects compare a firm to itself in election and nonelection years. Year effects absorb common world shocks. The remaining identification comes from the fact that different countries hold elections in different years.

2.3 Political and institutional variables

- **Close election:** an election whose victory margin is below the 25th percentile of the victory-margin distribution for the relevant sample.
- **Checks:** the number of veto players or checks on executive authority from the Database of Political Institutions.
- **Government stability:** an ICRG measure capturing the stability of the incumbent government.
- **Government/GDP:** ratio of central government spending to GDP.
- **Central bank independence:** an index from Cukierman et al. and Polillo–Guillen measuring how independent monetary authority is from politics.

Interpretation: These variables help distinguish a generic calendar effect from true policy uncertainty. The election effect should be larger when elections have more potential to change policy outcomes.

2.4 Cash-holding variables

- Cash holdings are modeled jointly with investment.
- Cash regressions control for firm size, leverage, cash-flow volatility, dividends, investment, country fixed effects, and year effects.
- The goal is to test whether the investment not undertaken in election years appears as higher cash holdings.

Interpretation: The cash analysis is important because it helps separate delayed investment from declining firm fundamentals. If firms simply became weaker in election years, cash need not rise.

3 Empirical specifications

3.1 Baseline investment regression

The baseline regression is

$$I_{ijt} = \alpha_i + \beta_1 Election_{jt} + \beta_2 Q_{i,t-1} + \beta_3 CF_{it} + \beta_4 \Delta GDP_{j,t-1} + \gamma_t + \varepsilon_{ijt},$$

where i indexes firms, j countries, and t years.

Interpretation: The coefficient β_1 measures the average within-firm investment change in the year leading up to a national election, relative to nonelection years and net of investment opportunities and macro conditions.

3.2 Cross-country heterogeneity

The paper then interacts election years with country characteristics:

$$I_{ijt} = \alpha_i + \beta_1 Election_{jt} + \beta_2 (Election_{jt} \times X_{jt}) + \beta_3 X_{jt} + Controls + \gamma_t + \varepsilon_{ijt}.$$

Interpretation: A negative β_2 means the election-year investment reduction is stronger when the country characteristic X is high. This is the main way the paper tests whether uncertainty intensity varies with institutions.

3.3 Predictability and close elections

For election outcome uncertainty, the paper estimates

$$I_{ijt} = \alpha_i + \beta_1 Election_{jt} + \beta_2 (Election_{jt} \times Close_{jt}) + Controls + \gamma_t + \varepsilon_{ijt}.$$

Interpretation: The uncertainty hypothesis predicts $\beta_2 < 0$: close elections should induce a larger investment decline because firms face more uncertainty over who will control policy.

3.4 Post-election dynamics

To examine whether investment rebounds after uncertainty is resolved:

$$I_{ijt} = \alpha_i + \beta_1 Election_{jt} + \beta_2 (Election_{jt} \times Close_{jt}) + \beta_3 PostElection_{jt} + \beta_4 (PostElection_{jt} \times Close_{jt}) + Controls + \gamma_t + \varepsilon_{ijt}.$$

Interpretation: If firms delay projects rather than cancel them, investment should recover after elections, especially after close elections.

3.5 Cash-holding system

The joint investment-cash system is

$$I_{ijkt} = \beta_0 + \beta_1 ElectionDummy_{jt} + \beta_2 Q_{ik,t-1} + \beta_3 CF_{ijkt-1} + \beta_4 \Delta GDP_{j,t-1} + \varepsilon_{ijkt},$$

$$Cash_{ijkt} = \beta_0 + \beta_1 ElectionDummy_{jt} + \beta_2 Q_{ik,t-1} + \beta_3 CF_{ijkt-1} + \beta_4 Size_{ijk,t-1} + \beta_5 LEV_{ijk,t-1} + \beta_6 I_{ijkt} + \beta_7 \sigma(CF)_{kt} + \beta_8 DIV_{ijkt}$$

Interpretation: This system asks whether the investment decline is mirrored by cash accumulation. The near-offsetting magnitudes are strong evidence for temporary delay.

4 Identification strategy

4.1 Core variation

- Different countries hold elections in different years, generating cross-country/year variation in political uncertainty.
- Firm fixed effects absorb permanent differences in firm investment levels.
- Year fixed effects absorb common global shocks.
- Controls for Q , cash flow, and GDP growth address investment opportunities, internal finance, and macroeconomic conditions.

Interpretation: The identifying assumption is that, absent election-related uncertainty, firms in election countries would have followed similar within-firm investment trends to firms in nonelection countries after controls.

4.2 Exogenous timing subsample

- The key concern is that incumbents may call elections when the economy is strong or weak.
- The paper classifies countries with fixed election timing separately.
- Results remain strong in fixed-timing countries, where election dates are more plausibly predetermined.

Interpretation: This helps rule out reverse causality from investment or macroeconomic weakness to election timing.

4.3 Uncertainty-intensity tests

- Close elections generate larger investment declines in fixed-timing countries.
- Countries with weaker checks and balances, lower government stability, and larger government sectors have stronger investment cycles.
- Politically sensitive industries have somewhat stronger election-year responses.

Interpretation: These tests are important because they move beyond a simple election dummy. They show that the effect is largest where election uncertainty should matter most.

4.4 Alternative explanations and falsification tests

- The paper omits crisis countries and major economies and obtains similar estimates.
- A random election placebo produces no effect.
- Macroeconomic political business-cycle regressions do not show systematic election-year manipulation in government spending, money growth, interest rates, or inflation.
- Omitting politically connected firms does not remove the result.
- Results hold when the incumbent is not running for reelection, weakening incumbent-opportunism explanations.

Interpretation: The robustness tests show that the result is not driven by a few countries, crisis periods, connected firms, or placebo election cycles.

4.5 Remaining threats

- Firms may anticipate policy changes before the formal election year.
- Election outcomes may correlate with unobserved expected policy changes not fully captured by controls.
- Worldscope coverage is tilted toward listed firms and may not represent private firms.
- The design identifies average election-year investment cycles, not the precise policy channel firms fear.

Interpretation: The evidence is strongest for the existence of an election-related investment cycle and for uncertainty as the best-fitting mechanism; it is less able to isolate a single policy dimension such as taxes, regulation, or public spending.

5 Tables and figures

5.1 Table I: Political Systems and Election Type

Read: Table I classifies countries by legal origin, executive structure, election timing, election type, and number of elections used.

Interpretation: The table documents the institutional foundation of the empirical design. It shows that the sample includes substantial variation in political systems and election timing.

Economic message: The election variable is not a generic calendar marker; it is built from country-specific political institutions.

Table I
Political Systems and Election Type

This table presents political system and election characteristics for each of the 48 countries in our sample between 1980 and 2005. The number of observations indicates the total number of firm-year observations included in Thomson Financial's Worldscope database for each country in our sample. A country is classified as presidential (parliamentary) if the president (prime minister) is chief of state and head of government. A country is also considered parliamentary if a hereditary monarch is the chief of state while the prime minister is the head of government. Other rules determining the classification of executive authority are described in the Internet Appendix. A country is classified as having flexible timing if the national leader or legislative body has the option to call an election before the regularly scheduled election date. The last two columns report the type and number of elections utilized in the analysis for each country.

Country	Number of Observations	Legal Origin	Basis of Executive Legitimacy	Election Timing	Type of Elections	Number of Elections
Argentina	887	French	Presidential	Fixed	Presidential	4
Australia	12,226	English	Parliamentary	Flexible	Legislative	9
Austria	1,916	German	Parliamentary	Flexible	Legislative	7
Belgium	2,916	French	Parliamentary	Flexible	Legislative	7
Brazil	3,674	French	Presidential	Fixed	Presidential	2
Canada	15,599	English	Parliamentary	Flexible	Legislative	8
Chile	1,922	French	Presidential	Fixed	Presidential	4
Colombia	430	French	Presidential	Fixed	Presidential	4
Czech Republic	508		Parliamentary	Flexible	Legislative	3
Denmark	3,700	Scandinavian	Parliamentary	Flexible	Legislative	8
Finland	2,553	Scandinavian	Hybrid	Flexible	Legislative	6
France	14,692	French	Hybrid	Fixed	Presidential	4
Germany	14,472	German	Parliamentary	Flexible	Legislative	8
Greece	3,206	French	Parliamentary	Flexible	Legislative	7
Hungary	386		Parliamentary	Fixed	Legislative	4
India	5,080	English	Parliamentary	Flexible	Legislative	6
Indonesia	3,003	French	Presidential	Fixed	Presidential	1
Ireland	1,386	English	Parliamentary	Flexible	Legislative	8
Israel	1,168	English	Parliamentary	Flexible	Prime Ministerial	3
Italy	5,025	French	Parliamentary	Flexible	Legislative	6
Japan	52,495	German	Parliamentary	Flexible	Legislative	9
Luxembourg	499		Parliamentary	Fixed	Legislative	4
Malaysia	8,301	English	Parliamentary	Flexible	Legislative	5
Mexico	1,801	French	Presidential	Fixed	Presidential	4
Netherlands	4,159	French	Parliamentary	Flexible	Legislative	8
New Zealand	1,451	English	Parliamentary	Flexible	Legislative	9
Norway	3,064	Scandinavian	Parliamentary	Fixed	Legislative	7
Pakistan	1,361	English	Parliamentary	Flexible	Legislative	5
Peru	762	French	Presidential	Fixed	Presidential	4
Philippines	2,119	French	Presidential	Fixed	Presidential	3
Poland	856		Hybrid	Flexible	Legislative	4
Portugal	1,373	French	Parliamentary	Flexible	Legislative	3

(continued)

Table I—Continued

Country	Number of Observations	Legal Origin	Basis of Executive Legitimacy	Election Timing	Type of Elections	Number of Elections
Russia	388		Hybrid	Fixed	Presidential	3
Singapore	4,952	English	Parliamentary	Flexible	Legislative	6
Slovakia	131		Parliamentary	Flexible	Legislative	2
South Africa	5,742	English	Parliamentary	Flexible	Legislative	5
South Korea	7,652	German	Hybrid	Fixed	Presidential	4
Spain	3,249	French	Parliamentary	Flexible	Legislative	7
Sri Lanka	269	English	Presidential	Flexible	Presidential	3
Sweden	5,097	Scandinavian	Parliamentary	Fixed	Legislative	7
Switzerland	4,504	German	Parliamentary	Fixed	Legislative	5
Taiwan	7,613	German	Hybrid	Fixed	Presidential	3
Thailand	4,595	English		Flexible	Legislative	7
Turkey	1,742	French	Parliamentary	Flexible	Legislative	3
United Kingdom	36,050	English	Parliamentary	Flexible	Legislative	6
United States	50,257	English	Presidential	Fixed	Presidential	7
Venezuela	376	French	Presidential	Fixed	Presidential	4
Zimbabwe	159	English	Presidential	Fixed	Presidential	2

(a) Table I, part 1

(b) Table I, part 2

5.2 Table II: Summary Statistics and Election-Year Investment Patterns

Read: Table II summarizes election frequency, political platforms, vote shares, firm characteristics, and investment around elections.

Interpretation: The unconditional investment rate falls from 0.0795 in nonelection years to 0.0727 in election years, an 8.5% raw decline. Panel D shows the drop is concentrated in the election year and rebounds afterward.

Economic message: The raw data already exhibit an election-cycle pattern before adding controls.

5.3 Table III: Baseline Investment Regressions

Read: Table III reports the baseline investment regressions. The election-year coefficient is negative across specifications and remains significant after firm and year fixed effects.

Interpretation: The main controlled estimate is around -0.0036 to -0.0041 . Because the dependent variable is capital expenditures over assets, this is an economically meaningful decline relative to average investment rates.

Table II
Summary Statistics

Panel A reports summary statistics for national elections held between 1980 and 2005 in 48 countries. Panel B reports summary statistics for the firm characteristics used in the analysis. Panel C reports summary statistics for investment rates in both election years and nonelection years, where the investment rate is defined as capital expenditures scaled by beginning-of-year book value of total assets. Panel D reports the annual mean investment rates around the elections. Year 0 indicates the year leading up to an election. See the Appendix for variable descriptions as well as the variable sources.

Panel A: Election Characteristics					
		Mean	Median	Std. Dev.	
Election Frequency (unit: year)		3.8	4.0	1.2	
Length of Term (unit: year)		4.4	4.0	0.8	
Political Platform of Government					
Market-Friendly (%)		63.3			
Left-Leaning (%)		36.7			
Percent of Votes Won in an Election					
Winner (%)		41.9	41.3	12.9	
Runner-up (%)		28.7	27.9	10.1	
Third place (%)		12.2	11.4	5.7	
Type of Elections					
Legislative (%)		76.2			
Presidential (%)		22.6			
Proportion of Elections with		45.6			
Exogenous Timing (%)					
Change of Government Head (%)		54.3			
Change of Ruling Party (%)		43.3			
Checks and Balances		3.95	4.00	1.79	
ICRG Government Stability Rating		7.82	8.00	1.91	
Central Bank Independence		0.42	0.40	0.19	
Government Spending/GDP		0.28	0.26	0.11	
Panel B: Firm Characteristics					
	<i>N</i>	Mean	Median	Std. Dev.	
Investment Rate (I_t/A_{t-1})	101,587	0.078	0.051	0.094	
<i>Q</i>	101,587	1.587	1.240	1.140	
Cash Flow	101,587	0.098	0.092	0.132	
Panel C: Mean Investment Rates in Election Years vs. Nonelection Years					
Election Years		0.0727	0.0473	0.0896	
Nonelection Years		0.0795	0.0523	0.0955	
Difference		-0.0067			
Diff (<i>t</i> -stat)		-9.74			
Panel D: Mean Investment Rates around Election Years					
Year	-2	-1	0	1	2
Investment Rate	0.0788	0.0796	0.0727	0.0802	0.0792

Economic message: Firms invest less before elections even after accounting for investment opportunities and macro conditions.

Table III
Baseline Investment Regressions

This table presents estimates from investment regressions of the type

$$I_{ijt} = \alpha_i + \beta_1 \text{Election Dummy}_{jt} + \beta_2 Q_{i,t-1} + \beta_3 CF_{it} + \beta_4 \% \Delta GDP_{j,t-1} + \gamma_i + \varepsilon_{ijt}.$$

where i indexes the firm, j indexes the country, and t denotes the year. The left-hand-side variable is capital expenditures scaled by beginning-of-year total assets. $Q_{i,t-1}$ is the proxy for Tobin's Q . CF_{it} is cash flow, and $\Delta GDP_{j,t-1}$ is the percentage change in real gross domestic product for a given country over the previous year. See the Appendix for variable definitions. Standard errors, clustered by country and year, are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Election Year Dummy	-0.0067 [0.0016]***	-0.0036 [0.0014]***	-0.0036 [0.0014]***	-0.0037 [0.0012]***	-0.0038 [0.0013]***	-0.0041 [0.0014]***
Q			0.0110 [0.0010]***	0.0056 [0.0012]***	0.0055 [0.0012]***	0.0041 [0.0010]***
Cash Flow				0.1885 [0.0165]***	0.1866 [0.0159]***	0.1737 [0.0117]***
GDP Growth					0.0365 [0.0192]*	0.0402 [0.0181]**
Constant	0.0794 [0.0053]***	0.0285 [0.0023]***	0.0289 [0.0022]***	0.0272 [0.0020]***	0.0263 [0.0021]***	0.0084 [0.0020]***
Observations	101,587	101,587	101,587	101,587	101,587	101,587
R^2	0.00	0.00	0.01	0.07	0.07	0.13
Fixed Effects	None	Firm Year	Firm Year	Firm Year	Firm Year	Firm Continent $\times$ Year

shocks to investment. The results are robust to the inclusion of these effects. The results from the baseline specification are consistent with the hypothesis.

(a) Table III: Baseline Investment Regressions

Table IV
Elections and Investment: Variation across Countries

This table presents estimation results for the following regression:

$$I_{ijt} = \alpha_i + \beta_1 \text{Election}_{jt} + \beta_2 \text{Election}_{jt} \times X_{jt} + \beta_3 X_{jt} + \beta_4 Q_{i,t-1} + \beta_5 CF_{it} + \beta_6 \% \Delta GDP_{j,t-1} + \gamma_i + \varepsilon_{ijt}.$$

where X_{jt} is a country-specific characteristic. The country characteristics variable is interacted with all the other independent variables to allow for differences across countries with regard to different country characteristics. *Common Law* is a dummy variable equal to one if the firm operates in a common law country, and zero if in a civil law country. *Presidential System* is a dummy variable equal to one if the type of election is presidential, and zero if legislative. See the Appendix for additional variable descriptions. Tobin's Q , cash flow, and GDP growth are included as control variables. Standard errors, clustered by country and year, are reported in brackets. *, **, and *** represent statistical significant at the 10%, 5%, and 1% level, respectively.

	Country Characteristic				
	(1)	(2)	(3)	(4)	(5)
	Common Law	Presidential System	Checks	ICRG Govt. Stability	Government/GDP
Election Year Dummy	-0.0087 [0.0033]***	-0.0068 [0.0016]***	-0.0066 [0.0013]***	-0.0066 [0.0026]**	-0.0042 [0.0022]*
Country Characteristic $\times$ Election Dummy	0.0041 [0.0022]*	0.0015 [0.0019]	0.0007 [0.0003]**	0.0004 [0.0002]**	-0.0114 [0.0036]***
Country Characteristic	0.0026 [0.0067]	0.0044 [0.0037]	-0.0001 [0.0004]	-0.0010 [0.0007]	-0.0006 [0.00094]
Constant	0.0523 [0.0101]***	0.0524 [0.0065]***	0.0301 [0.0031]***	0.0075 [0.0054]	-0.0087 [0.0077]
Observations	101,515	101,515	101,459	101,364	97,118
R^2	0.13	0.13	0.07	0.07	0.06
Fixed Effects	Industry Year	Industry Year	Firm Year	Firm Year	Firm Year

(b) Table IV: Variation across Countries

5.4 Table IV: Elections and Investment, Variation across Countries

Read: Table IV interacts election years with common-law origin, presidential systems, checks, government stability, and government spending over GDP.

Interpretation: The election effect is stronger when political systems have fewer checks, less stable governments, and larger government sectors. This supports the view that elections matter more when policy discretion is more consequential.

Economic message: Political institutions discipline the real effects of elections.

5.5 Table V: Real GDP Growth and Election Timing

Read: Table V compares GDP growth across election years, nonelection years, regular elections, and early elections.

Interpretation: GDP growth does not appear systematically lower in election years, which weakens the argument that low investment simply reflects poor macro conditions before elections.

Economic message: The investment cycle is not just a GDP-cycle artifact.

Table V
Real GDP Growth and Election Timing

The table reports the average percentage growth in real GDP relative to the previous year around various sample periods. Average GDP growth is reported in two ways. First, the equal-weighted average GDP assigns the same weights to each country-year. Second, the sample-weighted average assigns the same weights to each firm-year observation. The first column reports GDP growth over the full sample. The next two columns split the full sample into election years and nonelection years. The final two columns split the election year sample into regular election and early election years. An election is classified as regular if it is held within 6 months before or after the anticipated election date, which is calculated by adding the nominal term of the chief executive to the previous election date. Standard errors are reported in brackets.

Panel A: Equal-Weighted across Country-Years					
	Overall	Nonelection Years	Election Years	Regular Elections	Early Elections
Mean	6.11%	6.17%	5.97%	5.38%	7.01%
SE	[0.39%]	[0.47%]	[0.71%]	[0.95%]	[1.02%]

Panel B: Sample-Weighted Averages					
	Overall	Nonelection Years	Election Years	Regular Elections	Early Elections
Mean	4.70%	4.48%	5.51%	4.19%	6.98%
SE	[0.03%]	[0.03%]	[0.06%]	[0.09%]	[0.07%]

(a) Table V: Real GDP Growth and Election Timing

Table VI
Election Timing and Predictability of Outcomes

This table presents estimation results for the following regression:

$$I_{ijt} = \alpha_i + \beta_1 \text{Election}_{ijt} + \beta_2 \text{Election}_{ijt} \times \text{Close}_{ijt} + \beta_3 Q_{i,t-1} + \beta_4 CF_{it} + \beta_5 \% \Delta GDP_{j,t-1} + \gamma_i + \epsilon_{ijt},$$

where *Close* is a dummy variable set to one if the vote difference between the winner and the runner-up is below the 25th percentile of the victory margin distribution over the sample of countries under consideration. The first two columns present results for the sample of countries for which the timing of elections is fixed in time by electoral law. Columns (3) and (4) report the results for the sample of countries for which the government has the option to call the election early. The last two columns report the results separately for presidential and parliamentary political systems. Tobin's *Q*, cash flow, and GDP growth are included as control variables. Standard errors, clustered by country and year, are reported in brackets.

	Exogenous Election Timing		Endogenous Election Timing		Presidential	Parliamentary
	(1)	(2)	(3)	(4)	(5)	(6)
Election Year	-0.0041	-0.0021	-0.0035	-0.0032	-0.0025	-0.0036
Dummy	[0.0018]**	[0.0019]	[0.0015]**	[0.0011]***	[0.0022]	[0.0014]**
Election Year × Close Election		-0.0093		-0.0019	-0.0080	-0.0128
Constant		[0.0031]***		[0.0024]	[0.0026]***	[0.0054]**
	0.0249	0.0253	0.0416	0.0415	0.0498	0.0476
	[0.0014]***	[0.0020]***	[0.0062]***	[0.0066]***	[0.0061]***	[0.0052]***
Observations	45,897	45,897	55,690	55,690	30,225	71,519
<i>R</i> ²	0.08	0.08	0.06	0.06	0.09	0.08
Fixed Effects	Firm Year	Firm Year	Firm Year	Firm Year	Firm Year	Firm Year

(b) Table VI: Election Timing and Predictability of Outcomes

5.6 Table VI: Election Timing and Predictability of Outcomes

Read: Table VI shows that close elections in fixed-timing countries produce a much larger investment decline. The interaction between election year and close election is strongly negative in the exogenous-timing sample.

Interpretation: This is one of the strongest tests of the uncertainty channel. If firms are responding to electoral uncertainty, the effect should be larger when the outcome is harder to predict; this is exactly what appears in fixed-timing elections.

Economic message: The magnitude of the investment cycle rises with political uncertainty.

5.7 Figure 1 and Table VII: Investment around National Elections and Post-election Period

Read: Figure 1 displays the estimated investment cycle around fixed-timing elections. Table VII estimates post-election rebound regressions.

Interpretation: Close elections produce a sharper pre-election fall and a larger post-election rebound. The rebound is smaller than the pre-election decline, suggesting both temporary delay and some possible permanent real distortion.

Economic message: Firms wait before uncertain elections and partially resume investment after the uncertainty is resolved.

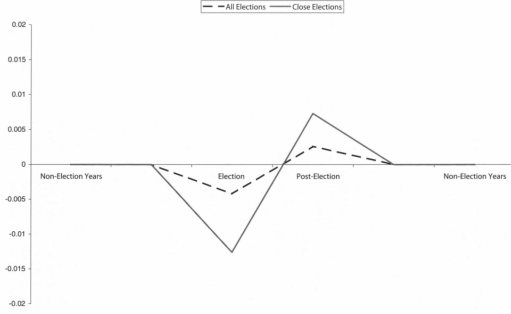


Figure 1. Investment around national elections. This figure displays estimates from the regression results reported in Table VII of the following specification:

$$I_{ijt} = \alpha_i + \beta_1 \text{Election}_{ijt} + \beta_2 \text{Election}_{ijt} \times \text{Close}_{ijt} + \beta_3 \text{Post-Election}_{ijt} + \beta_4 \text{Post-Election}_{ijt} \times \text{Close}_{ijt} + \beta_5 Q_{i,t-1} + \beta_6 CF_{it} + \beta_7 \% \Delta GDP_{j,t-1} + \gamma_i + \epsilon_{ijt}.$$

The sample includes only those countries for which the timing of elections is exogenous. The variable *Close* is a dummy variable set equal to one the margin of victory for a given election is smaller than the 25th percentile of the distribution for all exogenously timed elections. The vertical axis represents the investment rates relative to the average rate of non-election years, which are the periods neither immediately before nor immediately after an election. The dashed line displays changes in investment for all elections (based on estimates from column (2) of Table VII), and the solid line displays changes in investment for the elections with close outcomes (based on estimates from column (3) of Table VII).

(a) Figure 1: Investment around National Elections

Table VII
Changes in Investment in the Post-election Period

This table presents estimation results for the following regression:

$$I_{ijt} = \alpha_i + \beta_1 \text{Election}_{ijt} + \beta_2 \text{Election}_{ijt} \times \text{Close}_{ijt} + \beta_3 \text{Post-Election}_{ijt} + \beta_4 \text{Post-Election}_{ijt} \times \text{Close}_{ijt} + \beta_5 Q_{i,t-1} + \beta_6 CF_{it} + \beta_7 \% \Delta GDP_{j,t-1} + \gamma_i + \epsilon_{ijt}.$$

where *Close* is a dummy variable set to one if the victory margin, defined as the vote difference between the winner and the runner-up, is below the 25th percentile of the victory margin distribution across all elections for the sample considered. Tobin's *Q*, cash flow, and GDP growth are included as control variables. See the Appendix for additional variable definitions. Standard errors, clustered by country and year, are reported in brackets. The final two rows report the sum of the coefficients on the pre- and post-election variables along with the corresponding *t*-statistics for the null hypothesis that the coefficients of pre- and post-election variables sum to zero.

	Full Sample	Exogenous Timing Sample		Endogenous Timing Sample	
	(1)	(2)	(3)	(4)	(5)
Election Year Dummy	-0.0049 [0.0014]**	-0.0040 [0.0020]**	-0.0035 [0.0021]	-0.0034 [0.0013]**	-0.0032 [0.0014]**
Post-election Dummy	0.0030 [0.0019]	0.0028 [0.0022]	0.0029 [0.0023]	0.0045 [0.0029]	0.0049 [0.0030]
Election Year $\times$ Close Election			-0.0087 [0.0018]**		-0.0029 [0.0019]
Post-election $\times$ Close Election			0.0039 [0.0019]**		0.0025 [0.0017]
Observations	101,515	45,846	45,897	55,669	55,690
R^2	0.13	0.14	0.16	0.13	0.14
Fixed Effects	Firm Year	Firm Year	Firm Year	Firm Year	Firm Year
Tests for Linear Combinations of Coefficients					
Election + Post-election	-0.0019	-0.0012	-0.0054	0.0011	0.0013
<i>t</i> -statistic	-1.522	-1.071	-3.850	0.043	0.927

(b) Table VII: Changes in Investment in the Post-election Period

5.8 Table VIII: Additional Tests and Robustness

Read: Table VIII checks politically sensitive industries, incumbent ideology, crisis-country exclusions, lagged investment, random placebo elections, omission of the U.S./U.K./Japan, and equal country weights.

Interpretation: The election-year coefficient is robust except in the random-election placebo, where it disappears. The sensitive-industry interaction is negative, consistent with firms more exposed to government policy responding more strongly.

Economic message: The effect is not an artifact of a few large countries, regional cycles, or arbitrary election dates.

Table VIII
Additional Tests and Robustness

This table presents estimation results from various robustness tests. Tobin's *Q*, cash flow, and GDP growth are included as control variables. The random election dummy variable is a randomly generated election date for each country, calculated such that the number and periodicity of elections within each country is equal to that of the sample data. See the Appendix for additional variable definitions. The final column reports the average coefficients from country-by-country regressions of the investment specification. The standard error of the mean is included in brackets. For the remaining regressions, standard errors are clustered by country and year and reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Sensitive Industries	Incumbent Ideology	Crisis Countries Omitted	Lagged Investment	Random Election	U.S., U.K., Japan Omitted	Equal-Weighting by Country
Election Year Dummy	-0.0036 [0.0014]**	-0.0042 [0.0018]**	-0.0051 [0.0016]***	-0.0041 [0.0013]***	0.0001 [0.0004]	-0.0031 [0.0009]***	-0.0095 [0.0049]*
Sensitive Industry	0.0006 [0.0007]						
Election $\times$ Sensitive Industry	-0.0026 [0.0014]*						
Election $\times$ Market-Friendly Incumbent		-0.0053 [0.0023]**					
Observations	101,587	101,587	87,988	98,352	101,587	54,859	48
R^2	0.07	0.07	0.07	0.07	0.07	0.07	
Fixed Effects	Firm Year	Firm Year	Firm Year	Firm Year	Firm Year	Firm Year	Firm

Table VIII: Additional Tests and Robustness

5.9 Table IX and Table X: Political Connections, Incumbent Opportunism, and Political Business Cycles

Read: Table IX removes politically connected firms, examines high-central-bank-independence countries, and uses elections where the incumbent is not running. Table X tests macro political business-cycle variables.

Interpretation: The investment effect survives these tests, while macro variables such as government spending, money growth, real interest rates, and inflation do not exhibit systematic election manipulation.

Economic message: The evidence points more toward firm-level uncertainty than incumbent policy manipulation.

Table IX
Political Connections and Incumbent Opportunism

This table presents the estimation results of the baseline specification for various subsamples. The first column reports the estimation results of the investment specification omitting the politically connected firms from Faccio (2006). Column (2) reports estimation results for the subsample with a high degree of central bank independence (CBI). Specifically, the sample includes any observation in which the CBI measure is greater than the 75th percentile of the CBI distribution. The CBI index is defined according to Cukierman et al. (1992). The election year dummy in the final column is modified such that it is set to one only if the incumbent leader is not running for re-election in any given election, and zero otherwise. Tobin's *Q*, cash flow, and GDP growth are included as control variables. Standard errors, clustered by country and year, are reported in brackets.

	(1)	(2)	(3)
	Politically Connected Firms Omitted	High Central Bank Independence	Incumbent Not Running
Election Year Dummy	-0.0036 [0.0013]***	-0.0040 [0.0018]**	-0.0052 [0.0013]***
Constant	0.0262 [0.0020]***	0.0066 [0.0025]***	0.0247 [0.0021]***
Observations	98,137	43,322	101,587
<i>R</i> ²	0.07	0.06	0.07

(a) Table IX: Political Connections and Incumbent Opportunism

Table X
Political Business Cycle Regressions

This table reports the estimation results of political business cycle models based on panel regressions of time-series cross-section data for macroeconomic variables that take the general form

$$Y_{it} = \alpha_j + \beta_1 Y_{i,t-1} + \beta_2 Y_{i,t-2} + \beta_3 \text{World GDP Growth}_t + \beta_4 \text{Election}_t + \epsilon_{it}$$

where Y_{it} is the macroeconomic variable of interest and α_j is the country fixed-effect term. We report the regression results for annual growth in government spending (%ΔG), annual growth in the money supply (%ΔM1), the real interest rate (*r*), and the inflation rate (*i*). The first four columns report the results for the entire sample of countries. The final four columns report the regression results for only those countries with exogenous timing of elections. Macroeconomic data are obtained from the IMF Government Financial Statistics database and World Bank WDI data. Standard errors are clustered by country and are reported in brackets. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

	Full Sample				Exogenous Timing Sample			
	(1) %ΔG	(2) %ΔM1	(3) <i>r</i>	(4) <i>i</i>	(5) %ΔG	(6) %ΔM1	(7) <i>r</i>	(8) <i>i</i>
Election Year	0.0013 [0.0042]	-0.0164 [0.0148]	0.0072 [0.0045]	0.042 [0.1967]	0.0012 [0.0059]	-0.0357 [0.0284]	0.0055 [0.0035]	0.0609 [0.1222]
$Y_{i,t-1}$	-0.1367 [0.0824]	0.0463 [0.0191]**	0.5385 [0.0830]***	0.566 [0.3082]***	0.0458 [0.1087]	0.0225 [0.1005]	0.4679 [0.0575]***	0.618 [0.0582]***
$Y_{i,t-2}$	-0.1321 [0.0747]	-0.0198 [0.0359]	-0.0806 [0.0526]	-0.2816 [0.0689]***	-0.179 [0.1173]	0.0596 [0.0683]	0.0378 [0.0359]	-0.1913 [0.1176]
World GDP Growth	-0.3591 [0.3591]	3.0635 [1.3757]**	-0.364 [0.3266]	25.5446 [17.0890]	-0.3197 [0.6001]	1.4437 [2.0377]	-0.5148 [0.3093]	8.0223 [5.9885]
Observations	189	777	1,010	1,242	95	294	376	518
<i>R</i> ²	0.04	0.01	0.29	0.11	0.06	0.02	0.28	0.3

(b) Table X: Political Business Cycle Regressions

5.10 Table XI: Investment and Cash Holdings System of Equations Estimates

Read: Table XI estimates investment and cash holdings jointly. Investment falls by 0.0065 and cash holdings rise by 0.0053 in election years.

Interpretation: The cash increase nearly offsets the investment decline and the difference is not statistically significant. This is consistent with firms parking funds in cash while waiting for political uncertainty to resolve.

Economic message: The election-year investment decline looks like deferred investment financed internally, not simply a collapse in firm resources.

Table XI
Investment and Cash Holdings: System of Equations Estimates

This table reports estimates from the joint estimation of the following system of equations:

$$I_{ijt} = \beta_0 + \beta_1 \text{Election Dummy}_{jt} + \beta_2 Q_{ikt,t-1} + \beta_3 CF_{ijt} + \beta_4 \% \Delta GDP_{j,t-1} + \varepsilon_{ijt}$$

$$\text{Cash}_{ijt} = \beta_0 + \beta_1 \text{Election Dummy}_{jt} + \beta_2 Q_{ikt,t-1} + \beta_3 CF_{ijt} + \beta_4 \text{Size}_{ijt,t-1} + \beta_5 LEV_{ijt,t-1} + \beta_6 I_{ijt} + \beta_7 \sigma(CF)_{jt} + \beta_8 DIV_{ijt} + \alpha_j + \eta_{ijt}$$

where the right-hand-side variables in the investment equation are defined as previously, and the cash regression includes firm size (log of real total assets), leverage (book value of debt scaled by total assets), investment rates, cash flow volatility, dividend, and country fixed effects, $\sigma(CF)_{jt}$ is the time-series standard deviation of three-digit SIC cash flows calculated over the previous 4 years, DIV_{ijt} is a dummy variable set equal to one if the firm pays a dividend in year t and zero otherwise, and α_j captures country fixed effects. The estimation procedure is performed by three-stage least-squares full-information maximum likelihood estimation.

LHS Variable	Election Dummy	Q	Cash Flow	GDP Growth	Size	Leverage	Investment	CF Volatility	Dividend
Investment	-0.0065 [0.0007]***	0.0038 [0.0003]***	0.1895 [0.0022]***	0.0491 [0.0028]***					
Cash Holdings	0.0053 [0.0009]***	0.0286 [0.0004]***	-0.0489 [0.0034]***		-0.0009 [0.0004]***	-0.2235 [0.0073]***	-0.0498 [0.0044]***	1.0169 [0.0451]***	-0.0227 [0.0011]***
$H_0 : \beta_{elect}^{Inv} + \beta_{elect}^{Cash} = 0$									
$\chi^2(1)$	1.3600								
p-Value	0.2431								

Table XI: Investment and Cash Holdings System of Equations Estimates

6 Interpretive synthesis

- The paper documents a robust international investment cycle around elections.
- The strongest evidence for the uncertainty channel comes from close-election interactions, fixed-timing elections, institutional heterogeneity, and post-election rebounds.
- The cash-holding result ties the investment decline to real-options delay: firms do not simply shrink; they temporarily hold more cash.
- The evidence rejects several alternative political-business-cycle stories because macro policy variables show little systematic election-year manipulation.
- The paper's contribution is to show that political uncertainty can be a first-order determinant of corporate real activity, even in normal electoral democracies.

7 Short summary

- Firms reduce investment in election years by about 4.8% relative to normal investment.
- The effect appears in a large international sample of 48 countries and over 100,000 firm-year observations.
- The result is robust to firm and year fixed effects, firm-by-region/year effects, macro controls, and firm-level investment controls.
- The effect is stronger when elections are close and when institutions imply greater policy discretion or instability.
- Politically sensitive industries react more strongly, though the interaction is only marginally significant.
- Random placebo elections do not generate similar effects.
- Firms increase cash holdings during election years by nearly the amount they reduce investment.
- The evidence supports a political uncertainty channel rather than pure incumbent opportunism or macroeconomic manipulation.

8 Conclusion

8.1 Core conclusion

Julio and Yook show that national elections generate measurable cycles in corporate investment. The election-year investment decline is robust, economically meaningful, and concentrated in settings where uncertainty should be most important.

8.2 Economic intuition

Capital investment is partly irreversible. When elections may change future policy, firms can reduce downside risk by waiting. The value of waiting is larger when the election is close, when policy discretion is high, and when the industry is politically exposed.

8.3 What the paper adds

The paper extends political economy into corporate finance by showing that democratic political uncertainty affects firm investment decisions. It also adds evidence that the investment decline is linked to temporary cash accumulation, making the delay mechanism more credible.

8.4 Limitations

The paper does not identify the exact policy dimension firms fear, and the data cover listed firms rather than all firms. Election timing may still correlate with unobserved political expectations, especially outside the fixed-timing sample.

8.5 Takeaway

Political institutions shape corporate real activity. Elections are not merely political events; they can change the timing of investment, cash management, and the allocation of capital across time.